DATA7201 Data Analytics at Scale - Project Report

Examining Non-Government Aligned Election Spending

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Abstract

This report examines non-government advertisers on the Facebook ad platform around the May 2022 federal election in order to determine the existence of American-style political action committee (PAC) advertisers in Australian elections.

Facebook ad data surrounding the election was ingested and preprocessed using PySpark on YARN. A pandas UDF tagged candidate, party, and government advertising via token matching against the Australian Electoral Commission candidate roll. The remaining ads were clustered with Latent Dirichlet Allocation and hand-labelled by category to exclude non-political content.

The corpus was then analysed in two ways: a deterministic classifier based upon ad spending before and after the election, and a machine learning classifier using hashtags to create a training dataset for a logistic-regression text classifier. Neither metric alone was sufficient to properly classify advertisers.

Third-party advertising was found to be split into a number of distinct patterns, with political advertisers split into two groups: most or all of the ad spending before the election, or constant spending across the period with a pivot to election material before the election.

A number of advertisers were subsequently found who matched the PAC-style advertisers seen in the United States, including Climate 200, Smart Voting, GetUp!, ABC Friends, and the Australian Christian Lobby.

Contents

1 Introduction

Political Action Committees (PACs) are an American campaigning vehicle that raise money for political advertising outside of the existing political structure. Australia does not formally have PACs. Yet independent, third-party advertisers spend millions of dollars on advertising during Australian elections. Identifying these informal Australian “PACs” matters for political-advertising transparency, and for finding the boundary between advertisers who are campaigning on specific issues and advertisers who are campaigning on behalf of political parties with private funding.

The Facebook ad platform is attractive for advertisers as it provides access to a large viewer base with minimal effort — no door-knocking or physical advertising required. The Facebook Ad Library is also a convenient data source for answering this question. A dataset was provided covering four years of political advertising in Australia, from 2020 to 2024. Although this corpus covered a range of political events, this report focuses on the 2022 Australian federal election.

At approximately 320 GB of JSON files, the dataset cannot be analysed on a conventional workstation. A single machine cannot hold it in memory, and the joins required for deduplication and classification would be prohibitively slow. Distributed analytics frameworks are the natural fit. PySpark is well suited to this task because it distributes both storage and computation across a cluster, loads data lazily, and its DataFrame API supports efficient analysis across thousands of files. We also leverage Spark MLlib for both the unsupervised topic modelling (LDA) and the supervised text classifier (logistic regression).

The investigation was split into three steps: data ingestion and preparation, exploratory analysis, and election-window analysis. These are discussed in the following sections.

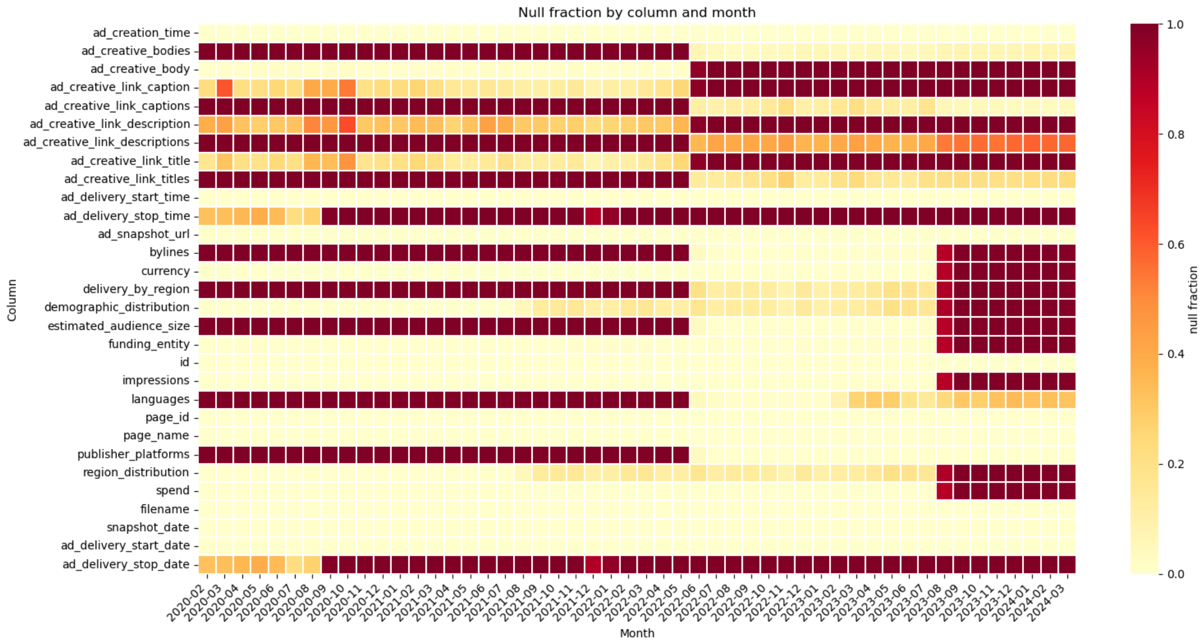
2 Dataset Analytics

2.1 Data ingestion and preparation

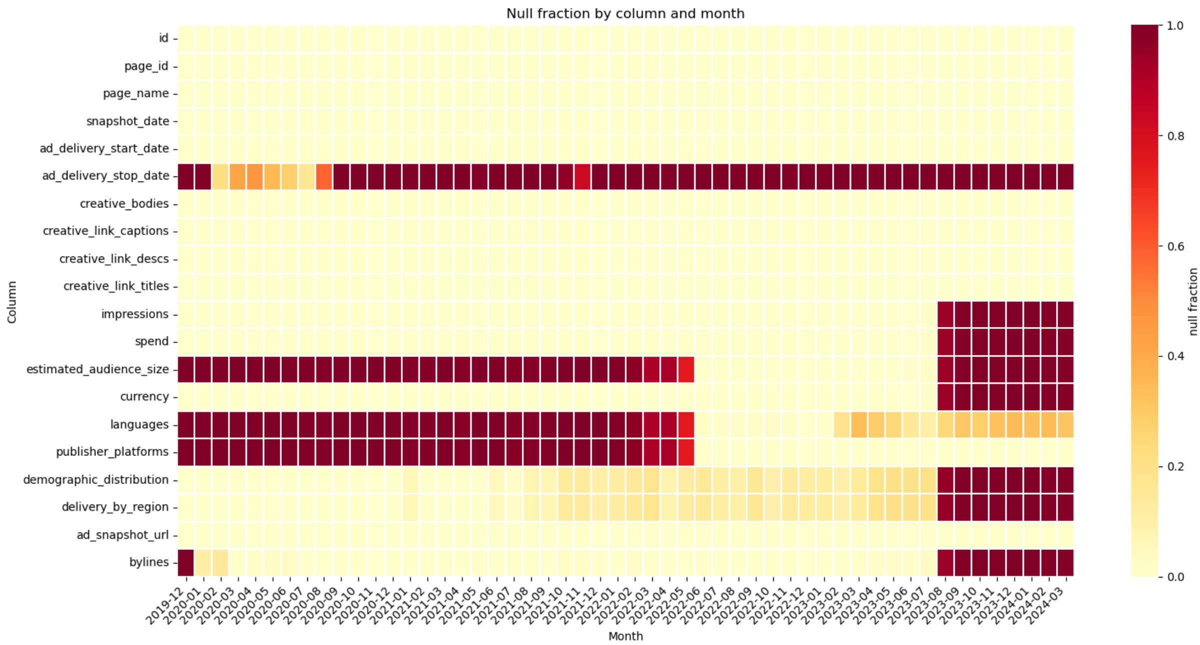
The data consists of approximately 4,500 JSON files containing around 40 million rows spanning February 2020 to March 2024. Each file is a snapshot of all Australian political advertisements active at that moment, so the same ad appears in multiple files. Each row represents a single ad with around 25 columns covering the ad purchaser, Facebook page, creative text, spend, impressions and other metrics. The schema varies over time, with a notable change in June 2022 and another in August 2023. Data becomes sparse in some key values after August 2023.

A null-fraction analysis was performed to visualise the schema drift. Automatic schema inference collected the superset of all fields across the dataset, and the per-column null fraction by month was plotted as a heatmap (Figure 1). Columns were then modified and merged to produce a consistent schema across the full date range. Various fields were stored as nested structs of string-typed lower and upper bounds; these were cast to numeric types and a midpoint was computed for each field, allowing downstream processes to work on a single value.

Deduplication was achieved by extracting the snapshot date from each filename and using it to sort snapshots chronologically, grouped by ad ID. A `row_number()` window operation tagged each row with an `ad_seq_no` where `1` indicates the most recent snapshot per ad. Filenames used inconsistent date conventions (some 7-digit, some 8-digit) and required special handling. The cleaned data was windowed to six months either side of the May 2022 federal election and written as Parquet to HDFS, reducing the corpus from approximately 40 million to 5.7 million rows.



(a) Before schema correction.



(b) After schema correction.

Figure 1: Null-fraction heatmaps (column × month) before and after the schema coalesce step. The raw view (top) shows the June 2022 singular-to-array transition and the August 2023 breakage clearly. The cleaned view (bottom) has the duplicate fields merged and a single consistent schema across the window. Hot colors show a high percentage of null values for that month/column.

2.2 Identifying government advertising

To focus on third-party organisations the government advertising had to be removed. This included both political-party advertisements and candidate advertisements. The 2022 House of Representatives and Senate candidate rolls were sourced from the Australian Electoral Commission. A straight join was not effective because page names can be misrepresentative and the byline can be paid by an aide. Thus a pandas UDF was used to apply three rules in order with the first match winning, summarised in Table 1.

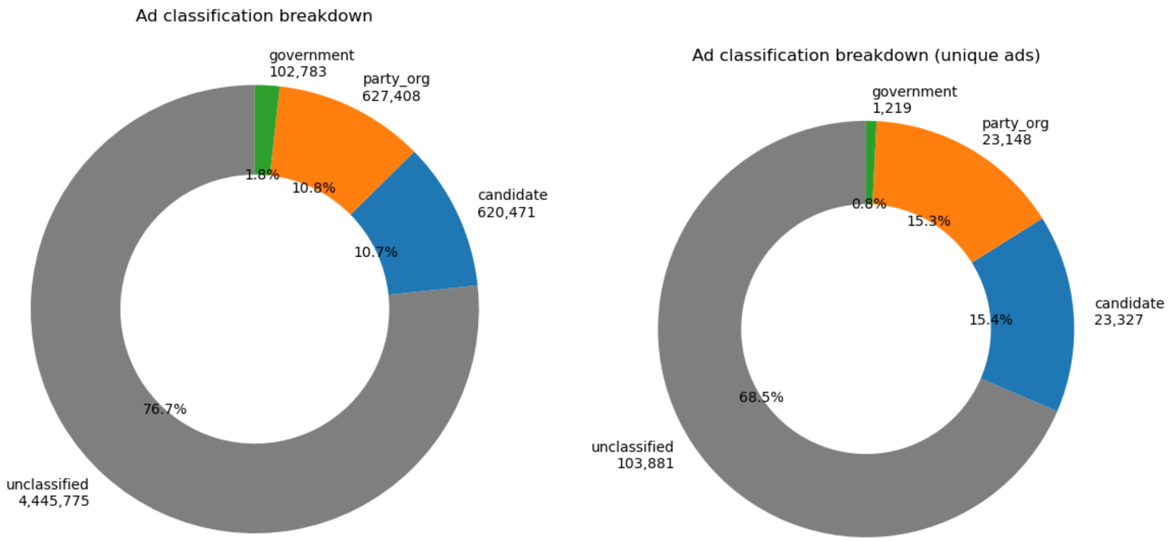
Rule	**Logic and output**
Candidate name	The candidate's tokenised name (surname plus first given name) is a subset of either the page-name tokens or the byline tokens.
Party byline signature	A tokenised party signature from `aec_parties.csv` is a subset of the byline tokens, e.g. {liberal, national} → LNP. Signatures are tested longest-first so LNP matches before LP.
Government byline signature	The byline tokens contain one of {department}, {australian, government}, or {australian, electoral, commission}.

Table 1: Three rules applied by the byline-matching pandas UDF in order, first match wins.

Every ad was thus tagged according to its nature; ads tagged `null` (no match against any of the three rules) formed the third-party corpus of interest. The row counts are shown in Table 2, with the match-type composition visualised in Figure 2 and the weekly spending pattern by match-type in Figure 3.

match_type	**total rows**	**unique ads**
null	4,511,742	105,814
party_org	649,857	23,932
candidate	536,232	20,644
government	98,660	1,188

Table 2: Rows and unique ads after government, party, and candidate classification.



(a) All rows (per snapshot).

(b) Unique ads (`ad_seq_no`=1).

Figure 2: Match-type distribution by total rows (left) and by unique ads after deduplication (right). The `null` bucket dominates by both measures and forms the third-party corpus carried forward.

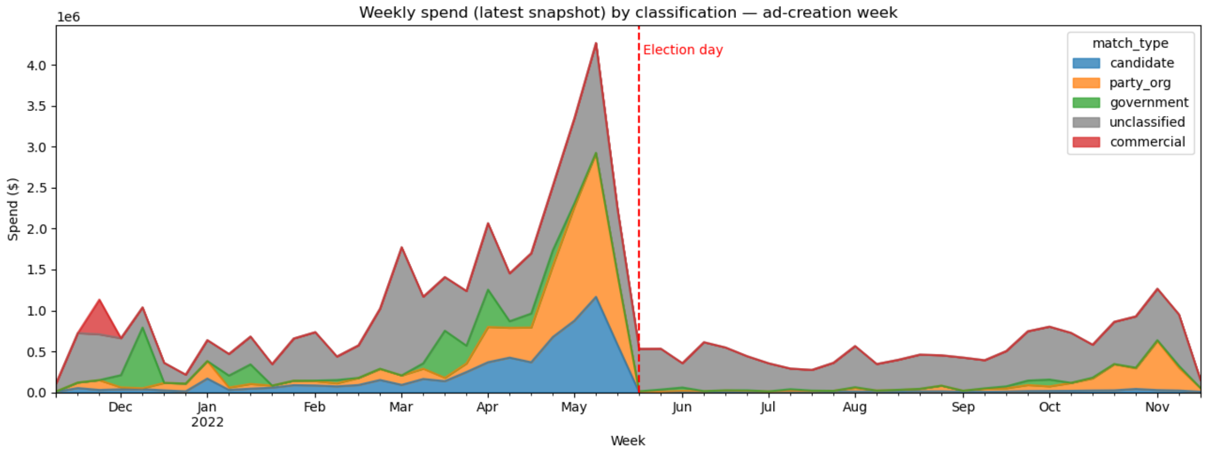


Figure 3: Weekly spend by match-type across the election window. The candidate and party advertising spike around the May 2022 election as expected; the `null` bucket shows a more diffuse pattern, which motivates the topic-clustering step in the next subsection.

2.3 Topic clustering via LDA

The non-government component consisted of a range of topics, not all relevant to the election. Latent Dirichlet Allocation (LDA) was used to cluster the non-government ads into topic groups based upon the ad text body. LDA is unsupervised, which was a significant advantage in a corpus of this size. Text was lowercased, tokenised, stop-words removed, and then vectorised using `CountVectorizer`. After tuning the number of topics k on a subset of the data, $k = 25$ was selected as it gave the cleanest separation. The topics were then hand-labelled with categories: `climate`, `humanitarian_rights`, `political_advocacy`, `cost_of_living`, and `mixed`.

Topics without a single coherent theme were tagged as `mixed` and kept in the corpus rather than dropped, so downstream analysis can choose whether to include them. The result was a corpus of about 95,000 advocacy ads, written partitioned by category for selective downstream reads. The per-topic ad distribution is shown in Figure 4.

Of significant interest for this analysis, the largest spending bucket was the non-government `political_advocacy` category.

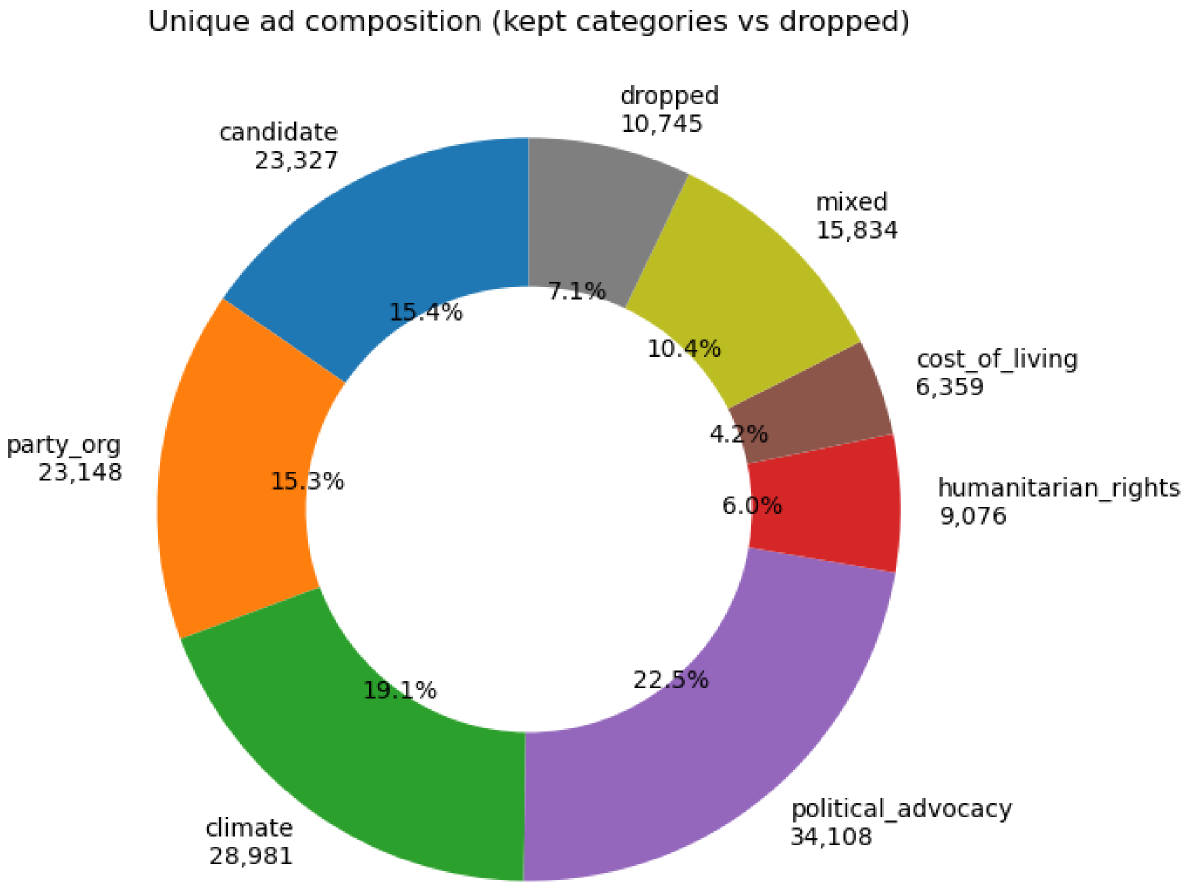


Figure 4: Distribution of ads across the 25 LDA topics with the hand-assigned categories.

2.4 Temporal analysis: persistence and centroid bands

A simple method for identifying "PAC"-style advertising is to look at spending before and after the election. An advertiser who has a high pre-election spending spike compared to the post-election period is more likely to be focused on the election.

In order to compare advertisers, spending was collated into a weekly time series per advertiser. As the time series was very bursty, a four-week rolling mean was then applied. A hard boundary at the election date was added to prevent bleed over from the pre-election month into the post-election month. The resulting weekly pattern across the non-government corpus is shown in Figure 5.

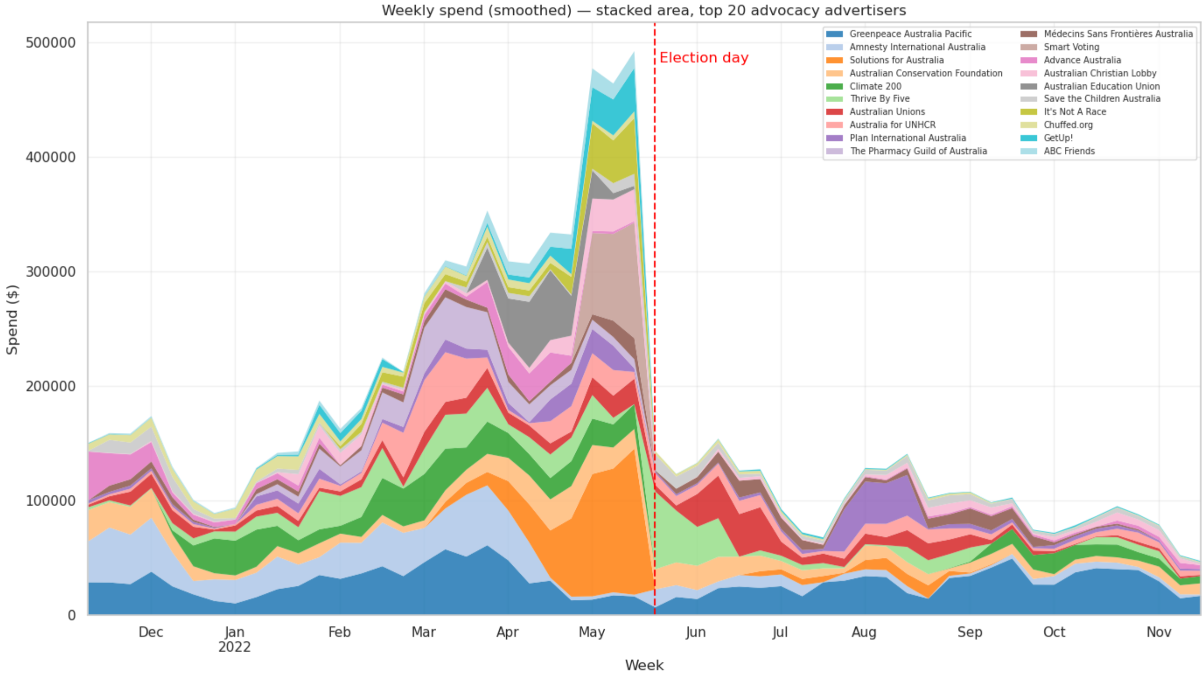


Figure 5: Weekly spend across the non-government (advocacy) corpus. Spending clusters in the weeks leading up to the May 2022 election and drops sharply afterwards.

A ratio of pre-election spend to post-election spend was calculated and saved as "persistence". Investigation of persistence found it was insufficient to identify strong pre-election spenders alone. Early-window spenders received the same weight as near-election spenders, and due to the crossover with the start of the Voice referendum, some political spenders had a late, post-election spend which reduced their ratio.

A second metric was thus developed. "Centroid" was the weighted-mean week from election day, where negative values indicate spending before the election and positive values indicate spending after. Centroid was then combined with persistence to identify advertisers who focused their spend in the near pre-election period.

Advertisers were split into four deterministic bands as shown in Table 3. The bands are also plotted on a quadrant in Figure 6.

Band	Rule
`election_only`	persistence < 0.30 and $\lvert\text{centroid}\rvert \leq 16$ weeks
`transitional`	$0.30 \leq$ persistence ≤ 0.80
`ongoing`	persistence > 0.80
`off_campaign`	persistence < 0.30 and $\lvert\text{centroid}\rvert > 16$ weeks

Table 3: Four-band classifier applied to top-20 advertisers.

Figure 6: Persistance vs Centroid Plot. Low persistance and centroid indicates ongoing spend. Larger persistance and a strong negative centroid indicates an advertisiting spend focused on the pre-election period.

2.5 Content analysis: hashtag-supervised classifier

Focusing on spend has some disadvantages. Firstly, it may not identify issue-based advertising that is not election-focused but incidentally had higher spend prior to the election. Due to a spike in spending around Christmas this was noted for a few advertisers. Secondly, the focus on spend misses advertisers who have ongoing spend but, during the pre-election period, target their spend on the election. Thus a content-based classifier was used to capture this.

Supervised classification requires a training corpus, and after some experiments, using hashtags was identified as a reliable method for extracting one. About 210 political hashtags such as `#federalelection` and `#morrisonmustgo` were extracted from the `election_only` category, and about 286 non-political hashtags such as `#breakfreefromplastic` and `#architecture` were extracted from the `ongoing` category. These hashtags were then hand-curated to remove any obvious false positives. A training dataset was then built from the relevant ads such that

- Positive: at least one political hashtag and no non-political hashtag.

- Negative: at least one non-political hashtag and no political hashtag.
- Ambiguous or unlabelled ads were excluded from training but scored at inference.

The hashtags were removed from the ad text prior to vectorisation so the model was forced to learn from the text body. A logistic regression classifier was then trained on this dataset. The training set contained 2,599 positive ads from 259 bylines and 5,262 negative ads from 303 bylines.

The held-out test AUC was 0.97 with 91% precision and 89% recall on the positive class. A leave-one-advertiser-out validation across the top 15 contributors gave an ad-weighted recall of 0.61. This is not a strong score, and suggests more advanced methodologies may be warranted. However, inspection of high-scoring no-hashtag ads showed the model correctly identified political content that the deterministic classification had missed, including Family First Party and several state-election candidates.

3 Discussion and Conclusions

The non-party, non-government advocacy ecosystem on Facebook around the 2022 election does not behave as a single block. The temporal classifier identifies four behavioural patterns.

The `election_only` band is the largest by count. It covers 11 of the top 20 advertisers, including Smart Voting, Solutions for Australia, Advance Australia, the Australian Education Union, GetUp!, It's Not A Race, Chuffed.org, Climate 200, and ABC Friends. These advertisers ramped up before the election and stopped immediately afterwards. Their cumulative spend curves elbow sharply at election day. Keyword analysis confirms that this group is dominated by election-specific language. This is the closest Australian analogue to US PAC-style temporal behaviour.

The `transitional` band contains advertisers that tapered rather than stopping dead. The `ongoing` band contains five year-round advocates whose post-election rate matches their pre-election rate. Greenpeace Australia dominates this tier at about $1.5M of spend across the window; the Australian Unions, Plan International Australia, Médecins Sans Frontières Australia, and Save the Children Australia each operate at roughly a third or less of Greenpeace's volume but maintain steady year-round messaging. No top-20 advertiser fell into the `off_campaign` band, since Shell was filtered upstream as a byline-tagged commercial advertiser in notebook 02.

The content classifier identifies a different signal. Among advertisers whose spend pattern looked steady, the content classifier reveals that some messaging pivoted toward electoral themes during the campaign period (defined as the four weeks before and after election day). The largest shifts are shown in Table 4.

Advertiser	**Outside campaign period**	**Campaign period**	**Shift**
Australian Christian Lobby	0.49	0.85	+0.35
Amnesty International Australia	0.35	0.51	+0.16
Médecins Sans Frontières	0.22	0.35	+0.13
Australia for UNHCR	0.18	0.27	+0.09
Plan International Australia	0.23	0.31	+0.08

Table 4: Top advertisers ranked by content-score shift between the campaign period and the rest of the analysis window. A positive shift indicates a content pivot toward electoral themes.

Greenpeace's content score dropped during the campaign period from 0.18 to 0.12. Their pre-election spending was focused on the Woodside gas campaign and not on electoral messaging. Climate 200's content score is high year-round at 0.78, so the campaign-period score of 0.85

shows no pivot because they were already maximally political. Advance Australia's content score moved the opposite way — 0.81 outside the campaign period versus 0.56 inside (−0.25). Their conservative culture-war messaging ramped up after the federal election rather than during it, which is consistent with the high-scoring Advance Australia ads dated October 2022 surfaced by the false-positive check (Victorian state campaign and pre-referendum content).

Neither metric alone separates the ecosystem cleanly. Temporal banding catches cliff-edge campaigners and off-cycle outliers but cannot detect content pivots in steady spenders. Content classification catches pivots but cannot speak to spend volume or timing concentration. Together they identify three patterns: PAC-style election-only campaigners visible to both methods, year-round advocates that pivot messaging during the campaign visible only to the content method, and year-round advocates that stay on issue visible only to the temporal method.

The dataset scale justified PySpark. The 40-million-row deduplication via window functions, the LDA fit on the residual corpus, and the pandas UDF for byline matching all leveraged Spark-specific features. None of these would have fit on a workstation in reasonable time. Hand-curation outperformed automation at two junctures: the LDA topic labels and the hashtag reference sets. Both were short enough for human review but provided strong supervision signal for downstream automation.

A Use of Generative AI

Claude (Anthropic; the `claude-opus-4-7` model accessed via the Claude Code CLI between April and May 2026) was used as a proofreading and editing assistant and as a coding agent during the preparation of this project. Its use is described below.

- **Writing.** The text of this report was written by the author. Claude's role in the writing was limited to proofreading and copy-editing: identifying typographical and grammatical errors, suggesting minor wording refinements, and flagging inconsistencies between paragraphs. All substantive content — the framing of the problem, the analytical argument, the choice of which findings to highlight, and the structure and narrative of the report — was authored and decided by the student.

- **Plotting.** Claude was primarily used as a teaching aide, using an interactive session to query issues rather than a search engine. Claude directly assisted with parts of the matplotlib visualisation code (donut charts, stacked-area plots, small multiples, and the quadrant scatter plots in notebooks 02, 04 and 05). The PySpark and pandas analysis pipeline was written by the author. All visualisation code generated with Claude's assistance was reviewed, tested, and integrated by the author.

The author retained responsibility for all analytical, code, and editorial decisions throughout, and verified the correctness of all code, numerical results, and claims before they were included in this report. Full conversation transcripts can be supplied on request.

B Code

The full analysis pipeline is implemented in six Jupyter notebooks at `https://github.com/<USER>/FB_API` (replace with the actual URL). Each notebook is self-contained and writes its output as Parquet on HDFS or as CSV in the repository's `data/` directory.

`01_data_loading.ipynb` Raw JSON ingestion, schema flattening, deduplication, window filtering, write of v1 Parquet.

`02_preprocessing.ipynb` Candidate, party, and government byline matching via pandas UDF; write of v2 Parquet.

03_topics.ipynb LDA topic modelling, k sweep, topic inspection, output of `topic_terms.csv` for hand-labelling.

04_topic_join.ipynb Join of hand-labelled topics onto v2, filter to political-adjacent residual, write of v3 Parquet.

04b_hashtag_analysis.ipynb Regex extraction of hashtags across v3, count by band, export of `data/hashtags.csv`.

05_election_spenders.ipynb Top-20 ranking, weekly aggregation, election-boundary smoothing, persistence + centroid computation, four-band classification, export of band CSVs and visualisations.

06_election_content_classifier.ipynb Hashtag-supervised logistic-regression text classifier; FP / FN / LOO validation; per-advertiser and per-week scoring; pivot-detection table.

Selected code excerpts follow.

Election-boundary smoothing (nb 05)

```
# Hard smoothing boundary at election day --- pre and post halves
# smoothed independently so cliff-edged advertisers do not bleed forward.
weekly['is_post'] = weekly['week'] >= ELECTION_DATE
weekly['spend_smooth'] = (weekly
    .groupby(['bylines', 'is_post'])['spend']
    .transform(lambda s: s.rolling(window=4, center=True, min_periods=1).mean())
       )
```

Hashtag-supervised labelling (nb 06)

```
pol_re = r'#(' + '|'.join(re.escape(t) for t in political_hashtags) + r')\b'
non_re = r'#(' + '|'.join(re.escape(t) for t in non_political_hashtags) + r')\b'

v3 = v3.withColumn('has_political_hashtag',     col('raw_text').rlike(pol_re))
v3 = v3.withColumn('has_non_political_hashtag', col('raw_text').rlike(non_re))

# Strip hashtags before vectorising --- model must learn from surrounding prose.
v3 = v3.withColumn('text', regexp_replace(col('raw_text'), r'#\w+', ' '))

labelled = v3.withColumn('label',
    when( col('has_political_hashtag') & ~col('has_non_political_hashtag'), 1.0)
    .when(~col('has_political_hashtag') &  col('has_non_political_hashtag'),
       0.0)
    .otherwise(None))
```